

MT 506: Biomaterials

Course Code	MT 506
Course Name	Biomaterials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech./PG./Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction and importance of biomaterials:** Overview of biomaterials and their critical role in modern medicine. Understanding the interdisciplinary nature of biomaterials, integrating biology, chemistry, engineering, and materials science. The impact of biomaterials in medical devices, implants, and tissue engineering (6 Hours)
- **Types of biomaterials:** Metallic Biomaterials: Stainless Steels, CoCr Alloys, Ti Alloys, Dental Metals, Other Metals, Corrosion of Metallic Implants, Manufacturing of Implants Ceramic Biomaterials: Nonabsorbable or Relatively Bio inert Bio ceramics, Biodegradable or Resorbable Ceramics, Bioactive or Surface-Reactive Ceramics, Deterioration of Ceramics, Bio ceramic Manufacturing Techniques Polymeric Biomaterials: Polymerization and Basic Structure, Polymers Used as Biomaterials, Sterilization, Surface Modifications for Improving Biocompatibility, Chemo gradient Surfaces for Cell and Protein Interaction Composite Biomaterials: Structure, Bounds on Properties, Anisotropy of Composites, Particulate Composites, Fibrous Composites, Porous Materials, Biocompatibility. (10 Hours)
- **Classification according to physiological response of biomaterials:** Bio inert, bioactive and bioresorbable biomaterials, Surface modifications, Surface analysis, Surface-protein interactions (4 Hours)
- **Biodegradable Polymeric Biomaterials:** Glycolide/Lactide Based Biodegradable Linear Aliphatic Polyesters, Non-Glycolide/Lactide Based Linear Aliphatic Polyesters. Non-Aliphatic Polyesters Type Biodegradable Polymers, Biodegradation Properties of Synthetic Biodegradable Polymers. Hydrolytic Degradation. Enzymatic Degradation, Surface vs. Bulk Erosion: (10 Hours)
- **Cell interactions with biomaterials:** General introduction to cellular structure and functions, Techniques and assays to determine the cell material interactions (Cytotoxicity Assays, Direct Contact Assay, Agar Diffusion Assay, Elution Assay, Adhesion/Spreading Assays, Migration Assays) (4 Hours)
- **Biomaterials implants and implementation problems:** Orthopaedic implants, artificial organs, dental materials, etc.; Implementation problems - inflammation, rejection, corrosion, structural failure. ((8 Hours)
- **Applications of biomaterials in various medical conditions:** Wound Healing and the Presence of Biomaterials, Biomaterials and Thrombosis. (4 Hours)

Laboratory Modules

1. NA

Text Books

1. NA

References

1. Johnna S. Temenoff, Antonios G. Mikos, **Biomaterials the Intersection of Biology and Materials Science**, Pearson, 2008. by Johnna S. Temenoff, Antonios G.
2. B. Rolando (Ed.), **Integrated Biomaterials Science**, Springer. 2002.
3. J.B. Park and J.D. Bronzino, **Biomaterials: Principles and Applications**, CRC Press. 2002.